

HW 4.1.2.b: Simplifying Trigonometric Expressions (More Practice)

Simplify each of the following trig expressions completely.

1.) $\frac{\tan^2 \theta + 1}{1 + \cot^2 \theta}$

2.) $\frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta}$

3.) $\sec \theta \cdot \tan \theta \cdot \cos \theta$

4.) $\sin^2 \theta \cdot \cot \theta \cdot \csc \theta$

5.) $\frac{1 - \cos^2 \theta}{\sin^2 \theta}$

6.) $\frac{\tan^2 \theta}{1 - \sec^2 \theta}$

7.) $\tan^2 \theta (\csc^2 \theta - 1)$

8.) $\frac{1 - \cos^2 \theta}{1 + \cos \theta}$

9.) $\frac{\cos^2 \theta}{1 - \cos^2 \theta}$

10.) $\frac{\sec^2 \theta - 1}{\tan \theta}$

11.) $\frac{\tan \theta + 1}{\sec \theta}$

12.) $\frac{1 + \cot \theta}{\csc \theta}$

13.) $\frac{\cos^2 \theta - 1}{\sin^2 \theta - 1}$

14.) $\cos \theta \cdot \csc \theta \cdot (\sec^2 \theta - 1)$

15.) $\frac{\sec \theta \cdot \tan \theta}{\tan^2 \theta + 1}$

16.) $\frac{\sin^2 \theta \cdot \cot^2 \theta}{1 - \sin^2 \theta}$

17.) $\frac{\cos^2 \theta - 1}{\cos^2 \theta \cdot \tan^2 \theta}$

18.) $\cos \theta (\sec \theta - \cos \theta)$

19.) $\cot \theta (\tan \theta + \cot \theta)$

20.) $\frac{\tan \theta + \cot \theta}{\cot \theta}$

21.) $\frac{\tan \theta}{\tan \theta + \cot \theta}$

22.) $\sec \theta \cot \theta - \cot \theta \cos \theta$

23.) $\sin \theta \tan \theta - \csc \theta \tan \theta$

24.) $\frac{\cot^2 \theta \cdot \cos^2 \theta}{\cot^2 \theta - \cos^2 \theta}$

25.) $\frac{\sin^2 \theta - \tan^2 \theta}{\tan^2 \theta \cdot \sin^2 \theta}$

26.) $\frac{(\sin \theta + \tan \theta)^2 + \cos^2 \theta - \sec^2 \theta}{\tan \theta}$

27.) $\frac{2 \sin \theta \cdot \cos \theta + (\sin \theta - \cos \theta)^2}{\sec \theta}$

28.) $\frac{\sin \theta (1 + \sin \theta)}{1 - \cos^2 \theta} - 1$

29.) $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} + 1$

30.) $\frac{\tan \theta - \tan \theta \cdot \sin^2 \theta}{2 \sin \theta \cdot \cos \theta}$

31.) $\frac{\sec \theta - \cos \theta}{3 \tan \theta \sin \theta}$

32.) $\frac{\sin^3 \theta + \cos^3 \theta}{1 - \sin \theta \cdot \cos \theta}$

33.) $\frac{\cos^3 \theta + \sin^3 \theta}{(\cos \theta + \sin \theta)^2}$

34.) $\frac{1 + \tan^2 \theta + \sec^2 \theta \cdot \cot^2 \theta}{\csc^2 \theta + \cot^2 \theta \csc^2 \theta}$

35.) $\frac{\cot \theta \cdot \sec^2 \theta - \cot \theta}{\sin \theta \cdot \tan \theta + \cos \theta}$

36.) $\frac{\cos \theta}{1 + \sin \theta} + \frac{\cos \theta}{1 - \sin \theta}$

37.) $\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta}$

Answers:

1.) $\tan^2 \theta$

2.) $2 \tan \theta$

3.) $\tan \theta$

4.) $\cos \theta$

5.) 1

6.) -1

7.) 1

8.) $1 - \cos \theta$

9.) $\cot^2 \theta$

10.) $\tan \theta$

11.) $\sin \theta + \cos \theta$

12.) $\sin \theta + \cos \theta$

13.) $\tan^2 \theta$

14.) $\tan \theta$

15.) $\sin \theta$

16.) 1

17.) -1

18.) $\sin^2 \theta$

19.) $\csc^2 \theta$

20.) $\sec^2 \theta$

21.) $\sin^2 \theta$

22.) $\sin \theta$

23.) $-\cos \theta$

24.) 1

25.) -1

26.) $2 \sin \theta$

27.) $\cos \theta$

28.) $\csc \theta$

29.) $2 \cos^2 \theta$

30.) $\frac{1}{2}$

31.) $\frac{1}{3}$

32.) $\sin \theta + \cos \theta$

33.) $\frac{1 - \cos \theta \cdot \sin \theta}{\cos \theta + \sin \theta}$

34.) $\tan^2 \theta$

35.) $\sin \theta$

36.) $2 \sec \theta$

37.) $2 \csc \theta$